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REQUIREMENTS FOR ACQUIRING THE PRODUCT(S) DESCRIBED HEREIN SHALL CONSIST OF THIS SPECIFICATION AND THE ISSUE OF THE FOLLOWING SPECIFICATION LISTED IN THAT ISSUE OF THE DODISS SPECIFIED IN THE SOLICITATION: MIL-B-7949

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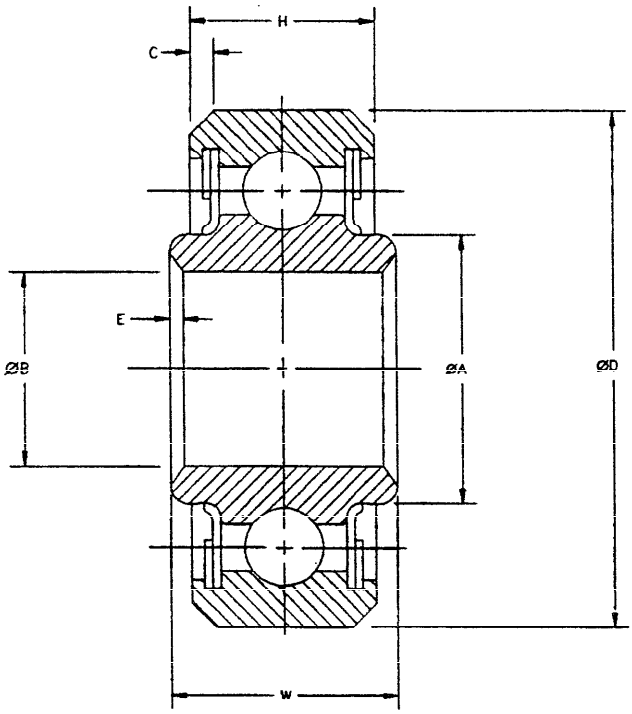


TABLE I. DIMENSIONS AND LOADS

MS DASH NO.	ØB 2/ BORE +.0000 −.0005	ØD OUTSIDE DIAMETER 1/ 2/ +.0000 −.0005	W WIDTH INNER RING 1/ +.000 −.005	H WIDTH OUTER RING 1/ +.000 −.005	ØA SHOULDER DIAMETER INNER RING (APPROX)	E 4/ INNER RING CORNER CHAMFER +.015 −.000	C 3/ OUTER RING CORNER CHAMFER +.015 −.000	RADIAL LIMIT LOAD RATING LBS	THRUST LIMIT LOAD RATING LBS	5/ 6/ RADIAL LOAD RATING (LBS.) FOR AVERAGE LIFE OF 10,000 COMPLETE 90° CYCLES CASE I	CASE II	WEIGHT POUNDS (APPROX)	MAXIMUM STARTING TORQUE (INCH-OZ)
-3A	.1900	.6250	.245	.203	.280	.005	.010	1560	700	1520	1260	.01	1
-3	.1900	.7774	.297	.270	.331		.022	1880	900	1700	1450	.03	1
-4	.2500	.9014	.484	.335	.390		.032	2680	1200	2410	2030	.04	1
-5	.3125	1.2500	.558	.375	.469	.015		5620	2500	4900	3970	.09	1
-6	.3750	1.4375	.620	.469	.591			7910	3500	6540	5410	.15	2
-8	.5000	1.6875	.620	.500	.768		.044	11800	5200	9320	7700	.21	3
-10	.6250	1.9375	.620	.500	.850			14100	6200	11000	9060	.28	3

1/ DIMENSIONS TO BE MET AFTER PLATING.
2/ OUT-OF-ROUND TOLERANCES: BORE: +.0002, −.0007;OUTER DIA: +.0005,−.0010.
3/ 45° CHAMFER OR A RADIUS GIVING APPROXIMATELY THE SAME GRIP FOR STAKING THE BEARING IN THE HOUSING WILL BE ACCEPTABLE.
4/ 45° CHAMFER OR A RADIUS GIVING APPROXIMATELY THE SAME FILLET CLEARANCE WILL BE ACCEPTABLE.
5/ CASE I − LOAD FIXED WITH RESPECT TO OUTER RING.
CASE II − LOAD FIXED WITH RESPECT TO INNER RING.
6/ THESE RATINGS ARE FOR OPERATION UP TO 250°F. FOR OPERATION UP TO 350°F, THE RATINGS SHALL BE REDUCED BY 20%.

Ⓕ DENOTES CHANGES

INCH-POUND

PREPARING ACTIVITY: NAVY-AS
CUSTODIANS: ARMY-AV NAVY-AS AIR FORCE-99 DLA-
REVIEW: ARMY-AR,AT,MI AF-84, DLA-IS ARMY-ME
PROJECT NUMBER: 3110-0926

MILITARY SPECIFICATION SHEET

TITLE
BEARING, BALL, AIRFRAME, ANTI-FRICTION, HEAVY DUTY

SPECIFICATION SHEET NUMBER
MS27640 21 JUNE 95 REV F

SUPERSEDING
MS27640E 05 DEC 94

AMSC- N/A FSC 3110

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REQUIREMENTS:

1. MATERIAL: RINGS: 52100 STEEL PER AMS6440, OR E52100 STEEL PER FED-STD-66.
   BALLS: 52100 STEEL PER AMS6440 OR 51100 STEEL PER AMS6449, OR E51100 OR E52100 STEEL PER FED-STD-66.
   SEALS: POLYTETRAFLUOROETHYLENE PER AMS3652 OR POLYTETRAFLUOROETHYLENE SHEET, GLASS FABRIC REINFORCED PER AMS3666.
   SEAL RETAINERS: ANY CORROSION RESISTANT STEEL.
2. LUBRICANT: MIL-G-81322 OR MIL-G-23827. ALL BEARINGS SHALL BE PACKED WITH GREASE CONFORMING TO MIL-G-81322 UNLESS OTHERWISE SPECIFIED. IF MIL-G-23827 IS REQUIRED, ADD THE LETTER "G" AFTER THE MS27640 DASH NUMBER. MIL-G-23827 SHALL NOT BE USED FOR OPERATION WHERE TEMPERATURES EXCEED 250°F.
3. HARDNESS: HEAT TREAT RINGS AND BALLS TO ROCKWELL "C" 60 TO 66 AND STABILIZE FOR OPERATION AT 250°F.
4. SURFACE ROUGHNESS: RACEWAYS AND BALLS SHALL BE 8 MICROINCHES Ra PER ANSI/ASME B46.1.
5. (F) PLATING: ALL EXTERNAL SURFACES EXCEPT BORE, SEALS AND SEAL RETAINERS SHALL BE ZINC-NICKEL IN ACCORDANCE WITH AMS 2417, TYPE 2, OR CADMIUM PLATED IN ACCORDANCE WITH QQ-P-416, TYPE I, CLASS 2, WITH A THICKNESS OF .0003 TO .0006 INCHES.
6. RADIAL INTERNAL CLEARANCE: WITHOUT "R" DESIGNATOR PART NUMBER INDICATES .0004 TO .0010
   WITH "R" DESIGNATOR PART NUMBER INDICATES .0002 TO .0005.
7. RADIAL ECCENTRICITY: INNER RING, .0010 MAX, OUTER RING, .0016 MAX.
8. LATERAL ECCENTRICITY: INNER RING, .0010 MAX, OUTER RING, .0016 MAX.
9. PART NUMBER: THE PART NUMBER SHALL BE SEQUENCED LEFT TO RIGHT WITH DESIGNATIONS IN THE FOLLOWING SEQUENCE:

EXAMPLE OF PART NUMBER:
MS27640-4RG = BEARING, BALL AIRFRAME, ANTIFRICTION, HEAVY DUTY, .2500 INCH BORE, .0002 TO .0005 INCH RADIAL INTERNAL CLEARANCE, FILLED 80% MINIMUM WITH MIL-G-23827, 250°F MAX.

- LUBRICANT DESIGNATOR (SEE LUBRICANT REQUIREMENTS)
- RADIAL INTERNAL CLEARANCE DESIGNATOR (SEE RADIAL INTERNAL CLEARANCE REQUIREMENTS)
- DASH NUMBER (SEE TABLE I)
- MILITARY SPECIFICATION SHEET NUMBER

NOTES:
1. UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS ARE IN INCHES.
2. THIS SPECIFICATION TAKES PRECEDENCE OVER ANY OTHER DOCUMENTS REFERENCED HEREIN.
3. REFERENCED DOCUMENTS ARE THE ISSUES IN EFFECT AT THE DATE OF INVITATION FOR BID.
4. THE PARTS COVERED BY MS20200 ARE CANCELLED. USE MS27640 FOR NEW DESIGN AND REPLACEMENT.
5. FOR MS27640-3A, METAL SHIELDS ARE ACCEPTABLE.

PREPARING ACTIVITY: NAVY-AS CUSTODIANS: ARMY- AV NAVY- AS AIR FORCE- 99 DLA- REVIEW: ARMY-AR,AT,MI AF-84, DLA-IS ARMY-ME PROJECT NUMBER: 3110-0926	MILITARY SPECIFICATION SHEET TITLE BEARING, BALL, AIRFRAME, ANTI-FRICTION, HEAVY DUTY	SPECIFICATION SHEET NUMBER MS27640 21 JUNE 95 REV F SUPERSEDING MS27640E 05 DEC 95 AMSC- N/A FSC 3110
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